

58. HIGH INTENSITY TORQUE TRAINING CAN INCREASE CYCLING PERFORMANCE

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Photo by Mattia Cioni on Unsplash.

In a previous [KisW Research Note](#), we saw that low cadence training did not increase performance more than the same training at free cadence. In that study, participants were amateur middle aged cyclists and low cadence intervals were performed at moderate intensity (zone 3 in a 5/7 zones model).

In [another just published KisW Research note](#), we saw that the % of maximum dynamic force produced during torque training (low cadence) can vary a lot depending on the exercise intensity. For example, **at the same rpm, when passing from the first threshold to VO2max, the % of the maximum dynamic force expressed on the pedals is almost doubled. This could have implications on fibres type recruitment and so training adaptations.**

To date, there is no data reporting a clear and unequivocal benefit of torque (low cadence) training (1). This could be due to (also) the fact that different combination of exercise intensity and cadence could lead to different outcomes, without making possible to generalize a conclusion about the effectiveness of torque training, without specifying the exercise intensity it is performed.

Regarding this, with the aim to keep the Knowledgeiswatt Community updated and open minded about the latest cycling research, here below you can find some hot of the press data published by Hebisz and Hebisz (Wroclaw University of Health and Sport Sciences, Poland) in 2024 on *Plos One*. (2)

In this study the authors tried to answer the following question:

Does high intensity (sprint and VO2max) intervals performed at low cadence (high torque) improve cycling performance more than the same training carried out at free cadence?

WHAT DID THEY DO?

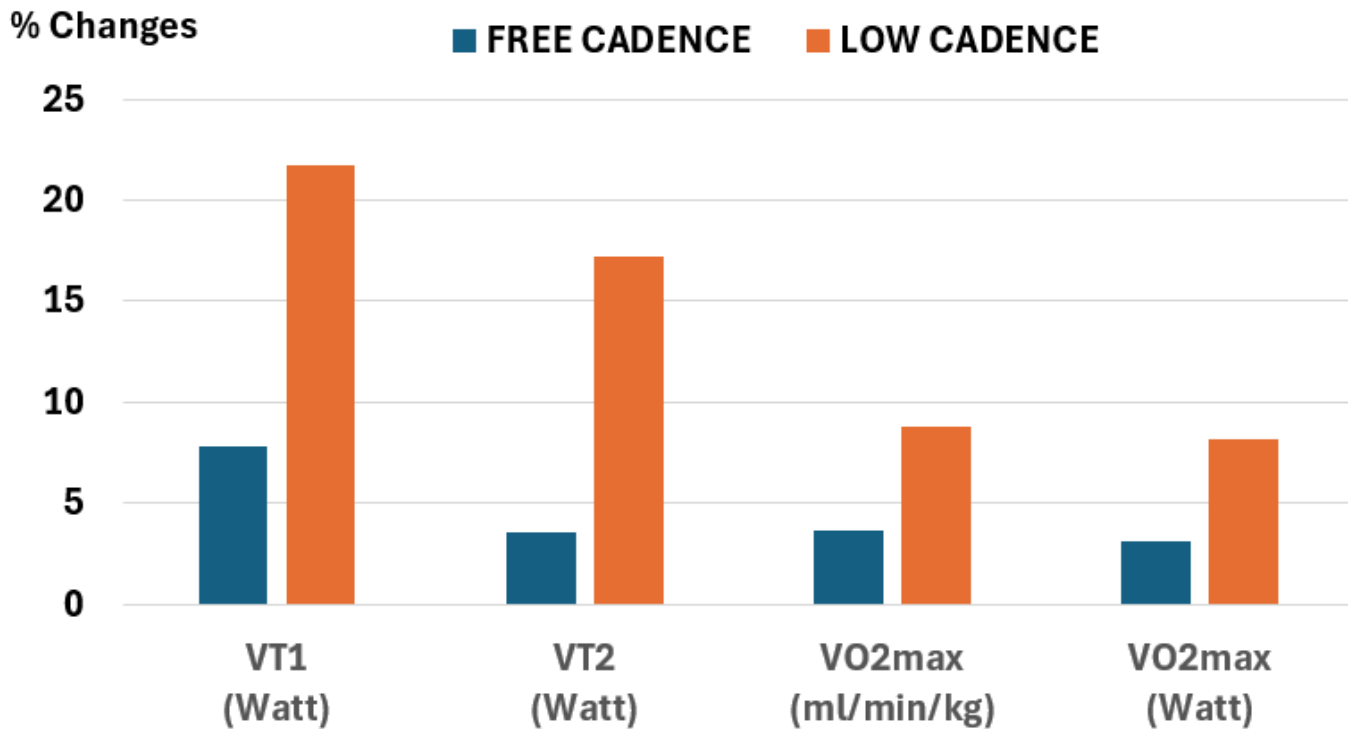
- 26 female, well-trained cyclists (age: 17-20, VO2max: 54) were divided into two groups (LOW CADENCE vs FREE CADENCE), which for 8 weeks performed both the following similar polarized training program:
 - 8 hours per week
 - 4 sessions per week:
 1. Sprints session: 2hrs with 8-12 x 30sec all-out sprints
 2. High Intensity (HIT) session: 2hrs with 4-6 x4min at ~110-120% of FTP power (zone 5 in a 5/7 zones model). 8min recovery between intervals.
 3. Long endurance session: 3hrs at an intensity close to first threshold
 4. Active recovery session: 1hour below the first threshold.
- The only difference between the two groups was the following:
 - LOW-CADENCE GROUP: performed the sprints and the intervals of the HIT sessions at 50-70 rpm.
 - FREE CADENCE GROUP: performed all the program at a freely chosen cadence (included the sprints and the HIT intervals) which was above 80 rpm.

WHAT DID THEY FIND?

Aerobic Physiological Parameters associated with cycling performance improved more in the LOW-CADENCE compared to the FREE-CADENCE GROUP:

VO2mas: +8% vs +4%

- First Threshold (VT1) Power: +22% vs +8%
- Second Threshold (VT2) Power: +17% vs +4%
- VO2max Power: +9% vs +4%



Data from Hebisz and Hebisz 2024. (2)

PRACTICAL TAKEAWAYS:

In young female cyclists, 8 weeks with 2 sessions per week including 30s sprints and VO2max intervals at low cadence (50-70 rpm) improve aerobic physiological parameters more than the same training performed at free cadence (>80 rpm).

GABRIELE'S COMMENT

This study is very interesting and add information on how torque training could be used to improve cycling performance. As said in the introduction, before this study was published, there were no good quality available data showing a superior effect of low cadence vs free cadence training to improve cycling performance.

So, why this study and not others showed that torque training is effective? I can suggest the 3 following possible reasons, but they are only speculation with no data to support. So, of course you are free to make your own interpretation of the data:

1. In this study, the intensity of low cadence training was very high (sprint and VO2max) when compared to the majority of other studies which did torque training below the second threshold power output. So, the relative force expressed during torque training here could have been (at least) twice higher than in other studies which showed no benefits.
2. Participants here were very young (17-20 years old) and so they could have been more responsive to high force training compared to middle-aged cyclists.
3. Participants here were female and so they could have been more responsive to torque training compared to male.

Anyway, when there will be more data published on torque/low-cadence training, we will publish them!

Thanks for reading! Share with your Team!

Gabriele Gallo, PhD in Exercise and Sport Sciences

REFERENCES:

1. Hansen EA, Rønnestad BR. Effects of Cycling Training at Imposed Low Cadences: A Systematic Review. *Int J Sports Physiol Perform*. 2017 Oct;12(9):1127-1136.
2. Hebisz R, Hebisz P. Greater improvement in aerobic capacity after a polarized training program including cycling interval training at low cadence (50-70 RPM) than freely chosen cadence (above 80 RPM). *PLoS One*. 2024 Nov 13;19(11):e0311833.



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